# Final experiment and evaluation addendum

Team Ultron | NLP Project 2 | CASIE event-conditioned argument extraction

This addendum extends the earlier five-model team report with four additional trained runs and a common nine-row test evaluation. The executed Colab notebook is the primary record of code, outputs and plots. The earlier report remains useful for its detailed five-model analysis; the final table below supersedes its model lineup.

## Task and data

Given an annotated cybersecurity event subtype and trigger, predict argument spans and one of 26 semantic roles. This is event-conditioned argument extraction with 53 BIO labels for the standard sequence models. It does not predict triggers.

| Split | Documents | Event mentions | Arguments |
|---|---|---|---|
| Training | 799 | 6,877 | 18,445 |
| Validation | 100 | 869 | 2,140 |
| Test | 100 | 736 | 2,109 |

The corrected 999-document corpus uses text-only tokenization, annotation alignment and a document-level split. The notebook verifies checksums, split isolation and the known near-duplicate constraint. The team preprocessing source appears in an optional appendix and was not rerun end to end in the additional training session.

## Evaluation design

Primary metric: exact span-and-role micro F1 (Arg-C). A correct prediction must match the event, start token, inclusive end token and role. Micro F1 combines counts across events; macro F1 averages role-level F1 and exposes poor performance on less frequent roles. Token accuracy would be dominated by non-argument tokens.

**Precision** = TP / (TP + FP); **recall** = TP / (TP + FN); **F1** = 2PR / (P + R). The selected model was fixed by validation F1. The consolidated notebook uses 1,000 document-level bootstrap resamples for 95% intervals, keeping correlated events in the same document together.

The original team test predictions had already been seen before the extra experiments, so this test set is not fully pristine. We disclose this and do not tune the new models on test scores.

# Nine test rows on the shared corrected split

Numbers below were copied from the executed consolidated notebook and use the same span scorer. The supplied CRF had no validation prediction file.

| Method | Validation F1 | Test F1 | Source |
|---|---|---|---|
| Feature CRF (additional) | 0.5075 | 0.5188 | Checkpoint |
| CRF (supplied) | - | 0.5251 | Prediction file |
| BERT cased | 0.5261 | 0.5354 | Prediction file |
| BERT uncased | 0.5557 | 0.5696 | Prediction file |
| BiLSTM-CRF | 0.5535 | 0.5415 | Checkpoint |
| DeBERTa-v3-base | 0.5987 | 0.5933 | Checkpoint |
| Role-conditioned DeBERTa | 0.5838 | 0.5966 | Checkpoint |
| SecureBERT | 0.6183 | 0.6140 | Prediction file |
| RoBERTa (selected) | 0.6189 | 0.6196 | Prediction file |

**Selection result.** RoBERTa led validation at 0.6189 and achieved 0.6196 test F1. Its 95% document bootstrap interval is 0.5968 to 0.6441. SecureBERT was very close (0.6183 validation; 0.6140 test). This small gap does not establish a general advantage without a direct paired comparison and further seeds.

## What the additional methods tested

The BiLSTM-CRF supplies a non-pretrained neural sequence baseline. The extra feature CRF is a separate run of an existing architecture. DeBERTa-v3-base adds a newer transformer encoder. The role-conditioned adaptation names one role in the input and predicts O/B-Target/I-Target; it queries every permitted role at inference. Its validation recall was 0.722, compared with 0.631 for standard DeBERTa, but precision dropped from 0.569 to 0.490. Its validation F1 was therefore lower.

A separate earlier DeBERTa run scored 0.6171 on validation, but the checkpoint was lost at a runtime reset. The checkpoint-backed run above scored 0.5987 and generated the displayed test predictions. We do not substitute the earlier number for it.

A RoBERTa + CRF head was considered but not trained. A fair comparison would need a newly trained matching RoBERTa softmax run because the supplied RoBERTa checkpoint is unavailable. The table reports completed experiments only.

# Errors, context and reproducibility

## Where the selected model fails

| Gold argument distance from trigger | Recall | Correct / gold |
|---|---|---|
| 1 to 5 tokens | 0.748 | 906 / 1,211 |
| 6 to 10 tokens | 0.520 | 212 / 408 |
| 11 to 20 tokens | 0.436 | 136 / 312 |
| 21 to 50 tokens | 0.281 | 45 / 160 |
| 51+ tokens | 0.167 | 3 / 18 |

Attack-Pattern and Purpose have lower F1 than frequent roles. Individual scores for roles with fewer than ten gold examples are unstable. The executed notebook includes the full per-role plot, subtype table, error examples, role confusions and paired bootstrap output. The displayed 0.994 paired bootstrap win fraction compares selected RoBERTa against the additional standard DeBERTa, not against SecureBERT; it is not a p-value.

## Published benchmark context

TextEE evaluates CASIE in a related gold-trigger setting but uses a reported 701 / 149 / 149 document split, rather than our corrected 799 / 100 / 100. Its preprocessing and scoring details may also differ. Published scores establish research context; they do not form a directly comparable leaderboard for these results.

## Artifacts and limitations

The single executed Colab notebook preserves the model code, validation and test tables, graphs, and error analysis. The project input bundle preserves the corrected data and the original team prediction files. Separate ZIPs preserve the newly trained checkpoints. The team did not supply weights for its original five models.

A Colab runtime reset happened after the final evaluation but before the optional per-event evidence ZIP was exported. The saved notebook retains the executed scores and plots. Recreating that additional ZIP requires checkpoint restoration and inference, not new training. The raw preprocessing appendix was retained for provenance, but was not independently rerun end to end in this session.

## References

CASIE: https://github.com/Ebiquity/CASIE
TextEE: Huang et al., Findings of ACL 2024, https://aclanthology.org/2024.findings-acl.760/
TAGPRIME: Hsu et al., ACL 2023, https://aclanthology.org/2023.acl-long.723/